DSC 140A - Midterm 01 Review

Problem 1.

a) Suppose $f(\vec{x})$ is a **convex** function of $\vec{x}$ and that the gradient of f at the point $\vec{x}^{(1)} = (0, 5)^T$ is $(-3, 5)^T$.

Let $\vec{x}^* = (x_1^*, x_2^*)$ be the minimizer of f; you can assume that $\vec{x}^* \neq \vec{x}^{(1)}$, which implies that $f(\vec{x}^*) < f(\vec{x}^{(1)})$. True or False: it must be the case that $x_2^* \geq 5$.

Solution:

False as the gradient is the direction of maximal ascent. We would move in the direction of $(3, -5)^T$ and therefore $x_2^* \leq 5$.

b) Explain in your own words why convexity is such a desirable property for a loss function when using gradient descent. What can go wrong when the function is *not* convex? Your answer should address the concept of local vs. global minima.

Solution: A convex function has the key property that any local minimum is also a global minimum. This means gradient descent, which only uses local slope information to decide which direction to step, is guaranteed to converge to the global minimum, provided the step size is chosen appropriately. When a function is non-convex, it can have multiple local minima that are not global minima, and gradient descent may get "trapped" in one of these suboptimal valleys depending on where it starts.

Problem 2.

Let $f_1(x)$ be a convex function and let $f_2(x)$ be a non-convex function. Define the new function $f(x) = f_1(x) + f_2(x)$. True or False: $f(x)$ must be non-convex.

○ True

● False

Solution: For a simple counterexample, let $f_1(x) = 4x^2$ and $f_2(x) = -x^2$. Then $f(x) = 4x^2 - x^2 = 3x^2$ is convex.

If that example seems too contrived, consider $f_1(x) = x^4 + 3x^2$ and $f_2(x) = x^3$. Then $f(x) = x^4 + x^3 + 3x^2$, and the second derivative test gives us $f''(x) = 12x^2 + 6x + 6$, which is positive for all x.

Problem 3.

Recall that a subgradient of the absolute loss is:

$$\begin{cases} \text{Aug}(\vec{x}), & \text{if } \text{Aug}(\vec{x}) \cdot \vec{w} - y > 0, \\ -\text{Aug}(\vec{x}), & \text{if } \text{Aug}(\vec{x}) \cdot \vec{w} - y < 0, \\ \vec{0}, & \text{otherwise.} \end{cases}$$

Suppose you are running subgradient descent to minimize the risk with respect to the absolute loss in order to train a function $H(x) = w_0 + w_1 x_1 + w_2 x_2$ on the following data set:

x_1	x_2	y
2	5	6
1	2	3

Suppose that the initial weight vector is $\vec{w} = (0,0,0)^T$ and that the learning rate $\eta = 1$. What will be the weight vector after one iteration of subgradient descent?

Solution:

As $\text{Aug}(\vec{x})\cdot\vec{w} = 0$, $\text{Aug}(\vec{x})\cdot\vec{w} - y = -y < 0$. So, $\frac{\partial}{\partial \vec{w}} R(\vec{w}) = \frac{1}{n}\Sigma_{i=1}^{n} \frac{\partial}{\partial \vec{w}} L_{abs}$. Plugging in our data we have, $\frac{\partial}{\partial \vec{w}} R(\vec{w}) = -\frac{1}{2}(\begin{bmatrix}1\\2\\5\end{bmatrix} + \begin{bmatrix}1\\1\\2\end{bmatrix}) = -\frac{1}{2}\begin{bmatrix}2\\3\\7\end{bmatrix}$ After 1 iteration of gradient descent $\vec{w} = (0,0,0)^T + \frac{1}{2}(2,3,7)^T = (1, 3/2, 7/2)^T$.

Problem 4.

Let $X = \{\vec{x}^{(i)}, y_i\}$ be a data set of n points where each $\vec{x}^{(i)} \in \mathbb{R}^d$.

Let $Z = \{\vec{z}^{(i)}, y_i\}$ be the data set obtained from the original by standardizing each feature. That is, if a matrix were created with the i-th row being $\vec{z}^{(i)}$, then the mean of each column would be 0, and the variance would be 1.

a) Suppose linear predictors H_1 and H_2 are fit on X and Z by minimizing the MSE, respectively.

True or False: $H_1(\vec{x}^{(i)}) = H_2(\vec{z}^{(i)})$ for every $i = 1, \ldots, n$.

Solution:

True. In linear functions, changing input unit do not change the output.

b) Suppose that X and Z are both linearly-separable. Suppose Hard-SVMs H_1 and H_2 are trained on X and Z, respectively.

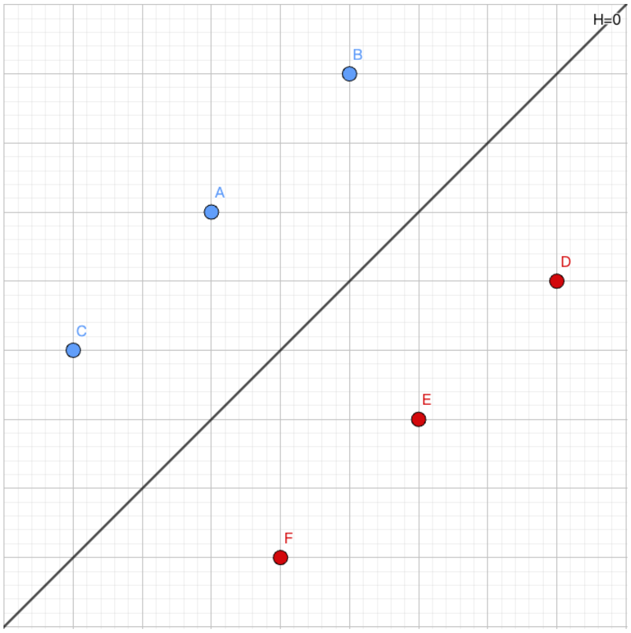
True or False: $H_1(\vec{x}^{(i)}) = H_2(\vec{z}^{(i)})$ for every $i = 1, \ldots, n$.

Solution:

False. In (Hard) SVMs, changing scales may change the output, e.g., by changing the set of support vectors.

Problem 5.

Consider the image below:



The blue points have label $+1$, and the red points have label -1. Suppose H is a linear prediction function, and when H is applied to the point A in the above image, $H(\vec{x}) = 4$. The black line in the middle of the image is the decision boundary $H = 0$.

You may assume that H is exactly in the middle of the points, and that all points are equidistant from the decision boundary.

a) What is the mean square loss of this prediction function, H?

Solution:

The output of H on the blue points is 4, while for the red points it is -4. This follows from the fact that all points are equidistant from the decision boundary, and H is linear. The target label for the blue and red points is 1 and -1, respectively. So the prediction is off by 3 in each case. This makes the squared error on each point $3^2 = 9$, so the mean squared error (mean square loss) is 9.

b) True or false: there exists a linear prediction function H which has a mean square loss of zero on this data.

Solution:

True. One simple way is to modify H. Let $H'(\vec{x}) = H(\vec{x})/4$, then H' is also a linear prediction function with its output exactly correct for each data point.

Alternatively, here's a geometric argument: a linear prediction function in two dimensions fits a plane to the data, where we have lifted to blue points to a height of 1 and the red points to a height of -1. Because the points are all equidistant from the black line in the figure above, there is a way to draw a plane that goes exactly through all of the points.

Problem 6.

Suppose a line of the form $H(x) = w_0 + w_1 x$ is fit to a data set of points $\{(x_i, y_i)\}$ in $\mathbb{R}^2$ by minimizing the mean squared error. Let the mean squared error of this predictor with respect to this data set be E_1.

Next, create a new data set by adding a single new point to the original data set with the property that the new point lies *exactly* on the line $H(x) = w_0 + w_1 x$ that was fit above. Let the mean squared error of H on this new data set be E_2.

Which of the following is true?

- ○ $E_1 < E_2$
- ○ $E_1 = E_2$
- ● $E_1 > E_2$

Solution: $E_1 > E_2$

Problem 7.

Consider a binary classification problem in $\mathbb{R}^2$ with labels $y \in \{-1, 1\}$. Suppose we train a linear classifier with weight vector $\vec{w} = (w_0, w_1, w_2)$ and use the predictor $\text{sign}(\text{Aug}(\vec{x}) \cdot \vec{w})$.

a) Explain in 2–3 sentences what the *decision boundary* is geometrically, and what role the *magnitude* of $H(\vec{x}) = \text{Aug}(\vec{x}) \cdot \vec{w}$ plays in a linear classifier's predictions. In particular, what does it mean for a correctly-classified point to have a large $|H(\vec{x})|$ vs. a small $|H(\vec{x})|$?

Solution: Since our prediction function is linear, the decision boundary is (geometrically) the plane where $H(\vec{x}) = 0$. The sign of $H(\vec{x})$ tells us which side of that hyperplane a point lies on, and hence which label we predict. The magnitude $|H(\vec{x})|$ is proportional to the *distance* of $\vec{x}$ from the decision boundary: a large $|H(\vec{x})|$ means the point is far from the boundary and the classifier is, in some sense, confident, while a small $|H(\vec{x})|$ means the point is close to the boundary and a tiny perturbation could flip the prediction.

b) Imagine a different a $H(x)$ that uses nonlinear features, for example $H(x) = \vec{w}^T \phi(x)$ where $\phi(x) = [x, x^2]^T$. Would our decision boundary still be a plane in our data space? Explain.

Solution: No, our decision boundary is not necessarily a plane. Because we have a nonlinear feature, our decision boundary (in our data space) can stretch and bend nonlinearly.

c) Suppose $\vec{w} = (w_0, w_1, w_2) = (1, 2, -1)$ (where $w_0 = 1$ is the bias). Consider the following four points and their true labels:

Point $\vec{x}^{(i)}$	True label y_i
(1, 3)	$+1$
(2, 1)	-1
$(-1,\ 2)$	$+1$
$(0,\ -1)$	-1

For each point, compute $H(\vec{x}^{(i)}) = \text{Aug}(\vec{x}^{(i)}) \cdot \vec{w}$, state whether it is correctly classified, and compute the **perceptron loss** ℓ_{tron} and the **0-1 loss** $\ell_{\text{0-1}}$ for each point.

d) The 0-1 loss directly measures what we care about (misclassification), yet we use other loss functions (like perceptron loss or hinge loss) during training. Why don't we use 0-1 loss?

Solution: Zero gradient almost everywhere. The 0-1 loss is piecewise constant: it equals 0 or 1 and is flat almost everywhere. Its gradient with respect to $\vec{w}$ is $\vec{0}$ almost everywhere, which means gradient (and subgradient) descent has no slope information to follow and cannot make progress.

Problem 8.

In class, we discussed how L1 regularization encourages sparse solutions and can be seen as a method for feature selection. In this problem, we will explore why L1 regularization promotes sparsity from the perspective of gradient descent.

a) First, write down the partial derivatives of the L1 and L2 regularization terms with respect to a specific weight w_j (you may ignore the case where $w_j = 0$, as the gradient might be undefined there).

- The L1 regularization term is given by:

$$R_1(\vec{w}) = \lambda \sum_{j=1}^{d} |w_j|$$

- The L2 regularization term is given by:

$$R_2(\vec{w}) = \lambda \sum_{j=1}^{d} w_j^2$$

b) Based on these derivatives, which regularizer is more effective at pushing w_j to zero?

Hint: Consider the behavior of the gradients when w_j is already small. For simplicity, assume that the partial derivative $\frac{\partial}{\partial w_j}$ of the Mean Squared Error (MSE) term is zero.

Solution: A) Partial Derivatives of L1 and L2 Regularization Terms

- **L1 Regularization:**

$$R_1(\vec{w}) = \lambda \sum_{j=1}^{d} |w_j|, \quad \frac{\partial R_1(\vec{w})}{\partial w_j} = \lambda \cdot \text{sign}(w_j)$$

  where $\text{sign}(w_j) = 1$ if $w_j > 0$ and -1 if $w_j < 0$.

- **L2 Regularization:**

$$R_2(\vec{w}) = \lambda \sum_{j=1}^{d} w_j^2, \quad \frac{\partial R_2(\vec{w})}{\partial w_j} = 2\lambda w_j$$

- **L1 Regularization** applies a constant force $\lambda \cdot \text{sign}(w_j)$, pushing weights to exactly zero and promoting sparsity.
- **L2 Regularization** applies a force proportional to w_j. When w_j is small, the gradient becomes small, making it less effective at driving weights to zero.

Problem 9.

Consider the following two scenarios:

- **Scenario A:** You apply the feature map $\vec{\phi}(x_1, x_2) = (x_1,\ x_2,\ x_1^2,\ x_2^2,\ x_1x_2)^T$ to your data and train a least squares classifier in feature space.

- **Scenario B:** You skip the feature map entirely and train a least squares classifier directly on $(x_1, x_2)^T$.

A classmate says: *"Scenario A is strictly better than Scenario B — more features always means a better model."* Another classmate disagrees: *"Scenario A is not always better. There are cases where adding these extra features could actually hurt you."*

Which classmate is correct? Describe a concrete situation where Scenario A would perform **worse** than Scenario B on new, unseen data. In your answer, explain the mechanism by which this happens.

> **Solution:**
>
> The second classmate is correct. While a richer feature map gives the model more expressive power, this is not always beneficial. Specifically, Scenario A can perform worse than Scenario B in the following situation:
>
> Suppose the true decision boundary between classes is actually linear in the original input space — for example, a straight line separating two well-behaved clusters. In this case, the extra features x_1^2, x_2^2, and $x_1 x_2$ provide no useful additional signal and the model has to learn to ignore these features.

Problem 10. (2 points)

Consider the data set: $\vec{x}^{(1)} = (1, 0, 2)^T$ $\vec{x}^{(2)} = (-1, 0, -1)^T$ $\vec{x}^{(3)} = (1, 2, 1)^T$ $\vec{x}^{(4)} = (1, 1, 0)^T$

Suppose a prediction function $H(\vec{x})$ is learned using kernel ridge regression on the above data set using the kernel $\kappa(\vec{x}, \vec{x}') = (1 + \vec{x} \cdot \vec{x}')^2$ and regularization parameter $\lambda = 3$. Suppose that $\vec{\alpha} = (-1, -2, 0, 2)^T$ is the solution of the dual problem.

a) What is the (2,3) entry of the kernel matrix?

1

> **Solution:** 1

b) Let $\vec{x} = (1, 1, 0)^T$ be a new point. What is $H(\vec{x})$?

> **Solution:** 14

Problem 11.

Let $\{\vec{x}^{(i)}, y_i\}$ be a data set of n points, with each $\vec{x}^{(i)} \in \mathbb{R}^d$. Recall that the solution to the kernel ridge regression problem is $\vec{\alpha} = (K + n\lambda I)^{-1}\vec{y}$, where K is the kernel matrix, I is the identity matrix, $\lambda > 0$ is a regularization parameter, and $\vec{y} = (y_1, \ldots, y_n)^T$.

Suppose kernel ridge regression is performed with a kernel κ that is a kernel for a feature map $\vec{\phi} : \mathbb{R}^d \to \mathbb{R}^k$.

What is the size of the kernel matrix, K?

- ○ $d \times d$
- ○ $k \times k$
- ○ $n \times n$
- ○ $n \times d$

> **Solution:**
>
> The kernel matrix is of size $n \times n$.

Problem 12.

In lecture, we saw that SVMs minimize the (regularized) risk with respect to the *hinge* loss. Now recall, the 0-1 loss, which was defined in lecture as:

$$L(\vec{x}^{(i)}, y_i, \vec{w}) = \begin{cases} 0, & \text{if } y_i = \text{sign}(\vec{w} \cdot \text{Aug}(\vec{x}^{(i)})), \\ 1, & \text{otherwise.} \end{cases}$$

a) Suppose a **Hard** SVM is trained on linearly-separable data. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x})$.

● True

○ False

Solution: True. If the data is linearly seperable, then a Hard-SVM will draw a decision boundary that goes between the two classes, perfectly classifying the training data and achieve a 0-1 risk of zero.

b) Suppose a **Soft** SVM is trained on linearly-separable data using an unknown slack parameter C. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x})$.

○ True

● False

Solution: False.

c) Suppose a **Soft** SVM is trained on data that is **not** linearly separable using an unknown slack parameter C. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x})$.

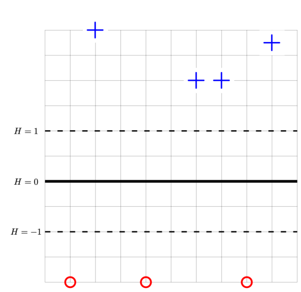
○ True

● False

Solution: False.

Problem 13.

The image below shows a linear prediction function H along with a data set; the "×" points have label +1 while the "∘" points have label -1. Also shown are the places where the output of H is 0, 1, and -1.



True or False: H could have been learned by training a Hard-SVM on this data set.

○ True

● False

Solution: False.

The solution to the Hard-SVM problem is a hyperplane that separates the two classes with the maximum margin.

In this solution, the margin is not maximized: There is room for the "exclusion zone" between the two classes to grow, and for the margin to increase.

Problem 14.

Consider the data set shown below:

x	y
-1	2
-3	3
2	4
4	7
5	7
6	10

What is the predicted value of y at $x = 3$ if the 3-nearest neighbor rule is used?

Solution: 6